

Juan David Muñoz Bolaños

Rechenauerstraße 42, Innsbruck, Österreich

+43 676 9115145

juan.munoz@i-med.ac.at

14. April 2026

ams-OSRAM AG

Frau Stefanie Kleierl

Tobelbader Straße 30

8141 Premstätten

Österreich

Bewerbung als Process Engineer Lithography (d/m/f) | Ref. 22816

Sehr geehrte Frau Kleierl,

Lithographie ist das Herzstück der Halbleiterfertigung und genau an der Schnittstelle von Physik, Optik und Prozesssteuerung fühle ich mich zu Hause. Als promovierter Physiker mit tiefgreifender Expertise in optischer Wellenfront Kontrolle, datenbasierter Prozessüberwachung und experimentellem Systemdesign bewerbe ich mich sehr gerne auf die Stelle als Process Engineer Lithography bei ams OSRAM in Premstätten.

Den direkten Bezug zur Halbleiterindustrie erwarb ich während der **ASML** und **ZEISS Semiconductor** Summer Schools, wo ich die physikalischen Grundlagen und industriellen Prozessherausforderungen moderner EUV DUV Belichtungssysteme sowie der Präzisionsoptik intensiv erarbeitete. Dieser Einblick in Belichtungstools, Overlay Kontrolle und CD Uniformität ergänzt unmittelbar meine Doktorarbeit: Ich habe vier Jahre lang **Wellenfrontaberrationen gemessen, analysiert und iterativ korrigiert**, exakt das physikalische Prinzip, das in der Lithographie über die Präzision von Focus und Overlay entscheidet. Mein selbst entwickelter Regelkreis entspricht methodisch der täglichen Überwachung und Stabilitätskontrolle einer Lithographie-Linie.

Die datenbasierte Prozesssteuerung ist ein weiterer meiner Kernbereiche. Im Rahmen meiner Promotion habe ich Mess und Steuerdaten aus Kameras, Wellenfrontsensoren und Deformable Mirrors in Python Pipelines ausgewertet, statistisch überwacht und zur automatischen Entscheidungsfindung genutzt. Am IPHT Jena entwickelte ich zudem das Python-Paket **SpectraMap** für die automatisierte Datenanalyse und Klassifikation hochdimensionaler Messdaten, eine Fähigkeit, die direkt in die datengetriebene **Yield-Optimierung** und Prozessüberwachung bei ams OSRAM einfließen kann. Meine Erfahrung in der Durchführung strukturierter Experimente (DoE), der statistischen Auswertung von Ergebnissen und der Ableitung konkreter Prozessverbesserungen ist durch drei peer reviewed Publikationen belegt.

Ich schließe meine Promotion im Juni 2026 ab, lebe seit vier Jahren in Österreich und freue mich darauf, den Sprung von der akademischen Forschung in die industrielle Fertigung bei einem globalen Technologieführer wie ams OSRAM zu machen. Mein Aufenthaltstitel wandelt sich nahtlos in eine Arbeitsbewilligung um. Ich freue mich sehr auf ein persönliches Gespräch.

Mit freundlichen Grüßen,

Juan David Muñoz Bolaños

Doktorand, Medizinische Universität Innsbruck

Juan David Muñoz Bolaños

Optical System Engineer

Rechenauerstraße 42, Innsbruck, Austria

+43 676 9115145

juan.munoz@i-med.ac.at | LinkedIn Profile



Summary

Physicist (PhD candidate) specializing in **optical wavefront control** and **data driven optimization**. Exposure to semiconductor lithography through **ASML** and **ZEISS SMT** training. Expertise in experimental system design and automated metrology. B2 German and Fluent English.

Professional Experience

Jan'22 | Present **Med. Univ. Innsbruck** Innsbruck, Austria
PhD Researcher | Process Control & Precision Metrology

Focus & Aberration Control: Designed a wavefront-sensorless closed-loop system for real-time aberration compensation and exposure focus control using an optofluidic DPP. Achieved a **3.5-fold signal-to-noise enhancement** and diffraction-limited precision in scattering environments. Published in *ACS Photonics* '25.

DoE & System Optimization: Conducted systematic **Design of Experiments (DoE)** and ZEMAX tolerance analyses to optimize high-order optical budgets. Applied iterative optimization algorithms to maintain sub-micron process stability.

Metrology Integration: Engineered and integrated transmissive metrology modules into complex R&D platforms; managed full technology transfer from prototype to stable daily clinical operation.

Lab Operations: 4 years managing a high-power ultrafast laser facility: EH&S compliance, high-precision equipment calibration, and technical documentation for industrial component procurement.

Jun'20 | Dec'21 **Leibniz IPHT** Jena, Germany
M.Sc. Researcher | Metrology & Data Analytics

Precision Metrology Design: Built a compact 1064 nm fiber-probe Raman spectrometer for deep-tissue process monitoring. Optimized NIR excitation for **yield-sensitive detection** and background reduction in complex chemical environments.

Statistical Process Control | SpectraMap: Developed a Python package for automated defect classification and **Statistical Process Control (SPC)** on high-dimensional hyperspectral maps. Published in *Applied Sciences* '24.

Mar'18 | Mar'19 **INTECOL S.A.S.** Medellín, Colombia
Junior Machine Vision Engineer | AOI Specialist

Automated Optical Inspection (AOI): Developed AI-driven machine vision algorithms for high-speed quality control. Engineered Halcon inspection pipelines and **Root Cause Analysis (RCA)** methodologies to detect labels and micro-defects on production lines. Achieved 100% customer sign-off in 3 iterations.

Technical Competencies

Lithography & Process Control: Direct exposure to DUV/EUV via **ASML/ZEISS SMT**; expert in **Focus & Overlay** budget management and active aberration compensation (*ACS Photonics '25*).

Yield Optimization & SPC: Statistical monitoring and automated defect classification (SpectraMap). Proficient in **Design of Experiments (DoE)** and root cause analysis (*Applied Sciences '24*).

System Metrology & AOI: End-to-end development of custom optical metrology tools and **Automated Optical Inspection (AOI)** pipelines (INTECOL) for high-speed manufacturing.

Software stack: Python (NumPy, SciPy, scikit-learn) for big data analysis, MATLAB for physical modeling, and C++ for real-time hardware control.

Education

Jan'22 | Jun'26
Austria

Medical Univ. of Innsbruck
PhD in Adaptive Optics & Precision Metrology

2019 | 2021
Germany

Friedrich Schiller Univ. Jena
M.Sc. in Photonics

2012 | 2018
Colombia

University of Cauca
B.Sc. in Physics Engineering

Publications & Awards

Awards: Selected for **ASML** (lithography) and **ZEISS SMT** (manufacturing technology). COLFUTURO 2026.

Muñoz Bolaños, J. D. et al. Confocal Raman Microscopy with AO. *ACS Photonics* (2025).

Muñoz Bolaños, J. D. et al. Dispersive 1064 nm fiber probe Raman. *Appl. Sci.* (2024).

Sohmen, M., Muñoz Bolaños, J. D. et al. Optofluidic AO in multiphoton microscopy. *Biomed. Opt. Expr.* (2023).

Soft Skills & Hobbies

Project Ownership: Sole developer of an AO microscopy system from optical design to publication (*ACS Photonics '25*); independently built the SpectraMap Python package (*Appl. Sci. '24*).

Interdisciplinary Teamwork: Collaborated with clinical and hardware teams across INTECOL, IPHT Jena, and Medical Univ. Innsbruck over 7+ years.

Systematic Problem Solving: Diagnosed and resolved systematic production defects at INTECOL; applied same methodology to wavefront control in published research.

Hobbies: Mountain biking, Bachata, Harmonica.

Languages

English (Fluent) · German (B2) · Spanish (Native)

References

Prof. Monika Ritsch-Marte
Medical University of Innsbruck
monika.ritsch-marte@i-med.ac.at

Dr. Christoph Krafft
Leibniz-IPHT Jena
christoph.krafft@leibniz-ipht.de

AGREEMENT FOR GUESTS

to carry out a practical training
with

Mr. **Juan David Munoz Bolanos** Date of birth: **1994-06-04**
temporarily living at **Emil-Wölk-Str. 7, 07747 Jena** (guest),

Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2020-06-02 up to 2020-12-31 admission to the necessary rooms within Leibniz-IPHT.

The use of further devices, media and other infrastructure takes place in agreement with the Heads of the respective departments.

Mr. Juan David Munoz Bolanos has to observe the security regulations for working and other regulations valid in Institute as well as instructions of the laboratory and clean room responsible persons of the Leibniz-IPHT.

Mr. Juan David Munoz Bolanos has to treat all affairs and procedures strictly confidential. Copies, duplicates and others of business and technical information of each kind are forbidden, unless these are sanctioned by the responsible Department Heads.

Publications, reports and lectures about topics which belong together with the fields of work of Leibniz-IPHT require the previous written agreement of the Executive Board.

The liability of Mr. Juan David Munoz Bolanos is limited to cases of wilful action and gross negligence.

Mr. Juan David Munoz Bolaos is insured against accidents on the Friedrich-Schiller-University Jena, Abbe School of Photonics in the context of his/her practical course during his/her length of stay at Leibniz-IPHT.

Jena, 18 May 2020


Prof. Dr. Popp
Chairman
of Leibniz-IPHT


F. Sonderrmann
Executive Member
of Leibniz-IPHT


Juan David Munoz Bolanos
Guest

AGREEMENT FOR GUESTS to carry out a practical training with

Mr. **Juan David Munoz Bolanos** Date of birth: **1994-06-04**
temporarily living at **Emil-Wölk-Str. 7, 07747 Jena** (guest),

Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2021-01-01 up to 2021-09-30 admission to the necessary rooms within Leibniz-IPHT.

The use of further devices, media and other infrastructure takes place in agreement with the Heads of the respective departments.

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Jena, 13 November 2020


Prof. J. Popp
Chairman
of Leibniz-IPHT


F. Sondermann
Executive Member
of Leibniz-IPHT


Juan David Munoz Bolanos
Guest

OFFICIAL TRANSLATION OF DOCUMENTS WRITTEN IN SPANISH, MADE 20
DECEMBER 2018 BY JUAN CARLOS OJEDA-GARCES, OFFICIAL TRANSLATOR
BY RESOLUTION No. 1171 ISSUED BY THE MINISTRY OF JUSTICE, REPUBLIC
OF COLOMBIA.



UNIVERSIDAD DEL CAUCA
ON BEHALF OF THE
REPUBLIC OF COLOMBIA
AN BY AUTHORIZATION OF THE MINISTRY OF NATIONAL EDUCATION
CONSIDERING THAT

JUAN DAVID MUÑOZ BOLAÑOS
CC 1.061.772.278 OF POPAYAN

HAS COMPLIED WITH ALL THE STATUTORY AND LEGAL REQUIREMENTS,
GRANTS THE TITLE OF

PHYSICAL ENGINEER

WITH ALL THE RIGHTS, PRIVILEGES AND DIGNITIES THAT AUTHORIZE HIM
FOR THE PROFESSIONAL EXERCISE

POPAYAN, 16 MARCH 2018

REGISTERED IN THE DIPLOMA BOOK No. 082 FOLIO 437 DIPLOMA No. 396
RESOLUTION No. R-266-07 MINUTE No. 14-07

(SIGNED)

THE PRINCIPAL OF THE UNIVERSITY
THE DEAN OF THE FACULTY
THE SECRETARY GENERAL OF THE UNIVERSITY

Juan Carlos Ojeda - Garces
TRADUCTOR e INTERPRETE OFICIAL
RESOLUCIÓN No 1171
MINISTERIO DE JUSTICIA 1997



URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

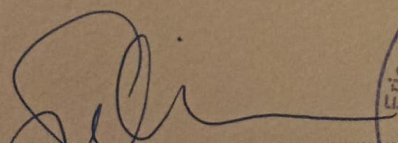
Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

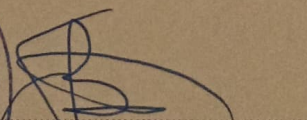
Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021


Prof. Dr. Christian Spielmann
Dekan




Prof. Dr. Silvana Botti
Vorsitzende des
Prüfungsausschusses

Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



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österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

Zertifikat

Juan David MUÑOZ BALAÑOS

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Mündliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 11.08.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ösd

österreich schweiz deutschland



ID-Nummer: ZB2Ö25m40434

Juan Munoz Bolanos

has participated in the

**ZEISS European
Summer School**

Lithography Optics &
Semiconductor Technologies

26 - 27 June 2025

ZEISS Semiconductor Manufacturing Technology, Oberkochen, Germany

Thomas Marschner

Dr. Thomas Marschner
Program Manager Funding Projects

Thomas Stammer

Dr. Thomas Stammer
CTO & Head of Product Strategy



ASML

PhD Master Class

Dear participant of the ASML PhD Master Class | Spring edition 2026,

We're looking forward to meeting you on March 19, 20 & 21.

Here's an information package containing details about the masterclass program. Please read this carefully so you know what to expect and how to prepare for it.

To get the most out of this masterclass for you, we want your input on what you would like to learn. At the end of this briefing, there is a link where we ask you to share your preferences with us.

Do not hesitate to contact us if you still have questions after this briefing.

Laura de Swart
Program Manager, ASML
+31 6 1562 7555

You will be joining us live at ASML HQ in Veldhoven on Friday and Saturday. Make sure you bring your **valid ID** to enter the ASML buildings. Without it, you will **not** be allowed inside. On Thursday we will meet at the Philips Stadium in the center of Eindhoven for an informal start to the event. The dress code for the entire masterclass is informal.

Program

Below is the program for the three days again. On the next pages, you will find more details on the different parts of the program.

PhD Master Class

Thursday, March 19

19:30 – 22:30 Informal activity @ Philips Stadium

Friday, March 20

08:30 – 09:00	Welcome
09:00 – 10:30	Introduction into ASML Technology
10:30 – 10:50	Break
10:50 – 12:00	Pitches from former participants
12:00 – 13:00	Lunch break
13:00 – 18:00	Working on cases
18:00 – 21:45	Dinner & presentations & feedback on case results
21:45 – 22:00	Closing day one

Saturday, March 21

08:30 – 09:00	Welcome back
09:00 – 12:50	Carousel • Matching conversations • HR corner • Q&A • ASML tours
12:50 – 13:50	Lunch break
13:50 – 14:00	Wrap up and closing

INTECOL

OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos (\$1,500,000) plus three hundred thousand pesos (\$300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL

NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO

Administrative and Financial Director

Integración Tecnológica Industrial Intecol S.A.S.
www.intecol.com.co
Medellin Calle 42 #63c-82
PBX. (574) 316 3322 Cell: (57) 301 430 2532

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Pasaporte N° / Passport No.

Tipo / Type

Cod. país / Country code

P

COL

Apellidos / Surname

PASAPORTE
PASSPORT

MUNOZ BOLANOS

Nombres / Given names

JUAN DAVID

Nacionalidad / Nationality

COLOMBIANA

Fecha de nacimiento / Date of birth
04 JUN/JUN 1994

Núm. personal / Personal No
CC1061772278

Sexo / Sex

Lugar de nacimiento / Place of birth

M

LA CRUZ COL

Fecha de expedición / Date of issue

11 DIC/DEC 2018

Fecha de vencimiento / Date of expiry

10 DIC/DEC 2028

Autondad / Authority

G. ANTIOQUIA

Firma del titular / Holder's signature

Dear Family

[illegible]